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# Palaeomagnetism and K–Ar Ages of Cenozoic Basalts from Argentina

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*Summary*

Results of a combined palaeomagnetic–geochronological study made on Cenozoic basalts from the extra-Andean area of the Provinces of Mendoza and Neuquén are presented and discussed in the light of the geological stratigraphy of the area and of the normal and reversed geomagnetic polarity intervals now well established for the past 4 My. The polarities of samples which yielded older dates are compared with the time zones of normal and reversed polarity but with arbitrary (i.e. not absolute) time scales, suggested by sea-floor spreading.

It is concluded that the volcanism during the Cenozoic in this area occurred in cycles. The most recent cycle occurred between the Upper Pliocene and the end of the Pleistocene in the north, centre and south of Neuquén province and in the south of Mendoza province. An older cycle extended from the Upper Miocene until the Middle Pliocene in the south and centre of Neuquén province.

We have defined new boundaries, which we have called G1 and G2, between normal and reversed epochs at $5{\cdot}5 \pm 0{\cdot}3$ My (R/N) and at $6{\cdot}6 \pm 0{\cdot}3$ My (N/R?) respectively.

## 1. Introduction

In recent papers, Valencio & Creer (1968) and Creer & Valencio (1969) presented the results of a palaeomagnetic study of Cenozoic lavas from the extra-Andean area of the Provinces of Mendoza and Neuquén. In the former paper were included the radiometric ages of some of these lavas obtained by the K–Ar method. The interpretation of this information allowed us to determine not only the characteristics of the geomagnetic field for this geological epoch but also threw new light on the stratigraphy of these lavas.

With a view relating the ages of these lavas to the geomagnetic polarity time scale, one of the authors (E.L.) dated 34 samples by the K–Ar method at the Geochronological Laboratory at the University of São Paulo, Brazil.

In this work the results obtained are presented with a critical discussion of the sources of error so as to compare the ages obtained with the stratigraphical ages and with the geomagnetic polarity time scale.

This is a preliminary study and it is intended to continue work on the dating of these Cenozoic lavas so as to improve the precision and coverage. Nevertheless, the data obtained up to now are considered sufficient to present here and to warrant interpretation.

## 2. Determination of radiometric dates

2.1.

About 60 g of sample were used of which 50 g were required for determination of radiogenic argon, whilst the finest fraction after sieving (150 mesh) was used for potassium determination.

The radiogenic $Ar^{40}$ content was determined by the method of isotopic dilution using the apparatus described by Amaral *et al.* (1966). About 15–20 g of sample were required and placed in an ultra-high vacuum and the system and sample degassed for 15–16 h. The sample was fused in a high frequency induction furnace and the gas obtained was mixed with a known volume of enriched $Ar^{38}$ ($Ar^{38}$ : 99·98 per cent), 'tracer'. Impurities were eliminated from the gas extracted in the high vacuum system by means of a zeolite trap and two furnaces, one of Cu+CuO and the other of Ti. The purified gas was transferred to a collector tube containing activated charcoal at liquid nitrogen temperature (−196 °C). The collector tube was sealed and transferred to the mass spectrometer for argon isotope analyses.

The isotopic analysis is performed on a gas-source mass spectrometer, of Reynolds type (Reynolds 1956), with a glass analyser tube of diameter 4·5 inches, magnetic field of 60°, and utilizing a static method of measurement.

When the gas is introduced the ratios $Ar^{40}/Ar^{38}$ and $Ar^{38}/Ar^{36}$ are determined by the spectrometer. This allows the calculation of the total quantity of argon–40 ($Ar^{40}_{tot}$) and of $Ar^{36}$ in the sample. The quantity of $Ar^{36}$ represents the content of atmospheric argon ($Ar^{40}_{atm}$) and allows the calculation of the contamination of the sample by argon from the atmosphere, thus:

$$Ar^{40}_{atm} = Ar^{36} \times 295{\cdot}6.$$

Then, the amount of radiogenic $Ar^{40}$ ($Ar^{40}_{rad}$) is given by:

$$Ar^{40}_{rad} = Ar^{40}_{tot} - Ar^{40}_{atm}.$$

The chemical analysis for potassium is made in the manner described by Brannock & Berthold (see Cordani 1968), using a Baird–Atomic photometer with a lithium internal standard.

Duplicate determinations of the amount of potassium are made, using in each case a blank and a standard. Thus the ambient contamination and that due to the chemicals used may be found, so that the error of measurement becomes 1 per cent or less.

2.2. *Calculation of age*

On the basis of the data for radiogenic $Ar^{40}$ and K, the age, $t$, is calculated as follows:

$$t = \lambda^{-1}.\ln(1+\lambda/\lambda_K.Ar^{40}/K^{40})$$

where

$$\lambda = 0{\cdot}530 \times 10^{-9}\,yr^{-1}$$

$$\lambda_K = 0{\cdot}585 \times 10^{-10}\,yr^{-1}$$

and

$$K^{40} = 0{\cdot}119\,K.$$

2.3. *Experimental errors*

In a good determination, the instrumental and analytical errors are considered to be of the order of 3 per cent, but may be larger in the cases of the rocks investigated here. This is due to the young ages, the amount of radiogenic argon being small. This is particularly so for basalts which contain only a small amount of potassium. The ratio of atmospheric to radiogenic argon is thus very high: the former may constitute 90 per cent of the total argon.

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This makes it difficult to correct for atmospheric $Ar^{40}$ so that the uncertainty in the determination of radiogenic $Ar^{40}$ can be large. Damon (1966), who studied this problem, made a graph which relates the percentage of atmospheric $Ar^{40}$ in the sample with the possible error in the determined age. For a contamination of about 50 per cent of atmospheric $Ar^{40}$, the error attributable to this source is lower than the experimental error so that the latter can be taken to be the error of the determined age. But for a sample containing respectively 90, 95 and 98 per cent atmospheric $Ar^{40}$, the said error can be 20, 40 and 100 per cent of the experimental error.

In the São Paulo geochronological laboratory, the instrumental error is about 3 per cent but since in almost all of our samples the percentage of atmospheric $Ar^{40}$ was greater than 50 per cent, we have preferred in all cases to estimate the error following Damon's studies because the error in the determination of radiogenic $Ar^{40}$ due to the presence of such large quantities of atmospheric $Ar^{40}$ is much greater than the instrumental errors.

The error could be reduced by dating several samples from a given lava flow.

### 2.4. *Results obtained*

In Tables 1 to 5 the results are given. The localities from which the samples were collected are given, the radiometric age, the magnetic polarity and the geological age. The locality map is given in Fig. 1 of Creer & Valencio (1969) which immediately precedes this paper.

## 3. Discussion of results

### 3.1. *The South Zapala area*

Here three basalt flows have been dated, all with reversed T.R.M. (Table 1). As was concluded from earlier geological studies, these three flows are Tertiary; nevertheless the one supposed to be Basalt II exposed at La Rinconada at the south of Catan Lil, gave a radiometric date appropriate to the Middle Miocene rather than to the Upper Pliocene. The data so far available suggest that in general the radiometric ages are older than those given by the geologists.

### 3.2. *Zapala area*

The results are summarized in Table 2, in which are also included some results already published for this area (Valencio & Creer 1968). In general, we observe again that the radiometric ages are older than those assigned to the same flows as a result of geological studies. The only exceptions are the Basalt I flows exposed to the south of Las Lajas (normal polarity, $4{\cdot}7 \pm 0{\cdot}5$ My and reversed polarity, $5{\cdot}4 \pm 0{\cdot}3$ My) and to the south of Zapala (normal polarity, $5{\cdot}7 \pm 0{\cdot}4$ My) which give a radiometric date appropriate to the Middle Pliocene rather than the Miocene. It is interesting to note that these dates confirm that the said flows can be considered to belong to a single effusive cycle. Moreover, two suggestive observations are noted: (a) the radiometric ages of these Basalts I, are less than those of the other flows from the same area catalogued as Basalts II and III; and (b) the radiometric ages of other Basalt I flows (Los Volcanes and South Zapala areas), are much greater than those of the similar basalts of this area. This last circumstance indicates a coherence in the radiometric dates which justifies a revision of the stratigraphic position of the Basalt I flows of this area.

In agreement with the radiometric study, the oldest lavas of the area are exposed in the Zapala plateau ($8{\cdot}0 \pm 0{\cdot}2$ My) and on road number 22 to the north of Zapala, at the altitude of the branch road to Estancia Llamuco ($8{\cdot}4 \pm 3{\cdot}5$ My), both of which

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**Table 1**

*Summary of the palaeomagnetic–geochronological data for South Zapala, Neuquén Province*

| | | K–Ar Method | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample No. | Locality | K per cent | $Ar^{40}$ rad mol. $x$ $10^{-11}$ $g^{-1}$ | $Ar^{40}$ atm. per cent | Age m.y. | Error m.y. | Magnetic polarity | Previous geological information Basalt No. | Age | New classification |
| 163–164 | 25 km south of Catan Lil on route No. 40 | 0·76 | 10·57 | 56·4 | 6·7 | ±0·3 | *R* | II | Ur. Pliocene | Lr. Pliocene |
| 165–169 | 37 km south of Catan Lil on route No. 40 | 0·69 | 52·24 | 40·4 | 16·0 | ±0·6 | *R* | II | Ur. Pliocene | M. Miocene |
| 154–157 | 113 km south of Zapala on route No. 40 | 1·01 | 16·49 | 45·8 | 20·6 | ±0·8 | *R* | I | Miocene | Lr. Miocene |

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## Table 2

*Summary of Palaeomagnetic–Geochronological Data for the Zapala Area, Neuquén Province*

| Sample No. | Locality | K–Ar Method: K per cent | $Ar^{40}$ rad mol. $x$ $10^{-11}$ $g^{-1}$ | $Ar^{40}$ atm. per cent | Age My | Error My | Magnetic polarity | Previous geological information: Basalt No. | Age | New classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 135–140 | Estancia Llamuco | 1·89 | 1·03 | 96·6 | 0·5 | ±0·3 | *N* | V | Lr. Holocene | Ur. Pleistocene |
| 100–113 | Portada de Covunco | 2·53 | 4·98 | 91·7 | 0·7 | ±0·2 | *N* | V | Lr. Holocene | Ur. Pleistocene |
| 202–209 | 35 km south of Las Lajas on road to Est. Llamuco | 1·89 | 5·78 | 89·6 | 1·3 | ±0·4 | *R* | IV | Ur. Pleistocene | Pleistocene |
| 141–144 | South of Est. Llamuco | 2·17 | 2·05 | 79·8 | 2·3 | ±0·3 | *N* | III | Lr. Pleistocene | Ur. Pliocene |
| LB 37 | Road to Est. Llamuco, 2 km to the west of route No. 22 | 1·44 | 8·58 | 57·4 | 2·6 | ±0·1 | *N* | III | Lr. Pleistocene | Ur. Pliocene |
| 172–177 | 20 km to S. of Las Lajas road to Zapala | 0·92 | 0·77 | 60·4 | 4·7 | ±0·5 | *N* | I | Miocene | M. Pliocene |
| 178–185 | 16 km to S. of Las Lajas road to Zapala | 1·37 | 16·49 | 57·0 | 5·4 | ±0·3 | *R* | I | Miocene | M. Pliocene |
| 152–153 | 13 km to S. of Zapala on route No. 40 | 0·75 | 0·76 | 63·9 | 5·7 | ±0·4 | *N* | I | Miocene | M.-Lr. Pliocene |
| 129–134 | Puesto Ramadilla, Est. Llamuco | 1·12 | 23·26 | 74·9 | 6·6 | ±0·3 | *N* | III | Lr. Pleistocene | Lr. Pliocene |
| 148–151 | 5 km to the S. of Zapala | 0·72 | 1·90 | 52·6 | 8·0 | ±0·2 | *R* | II | Ur. Pliocene | Ur. Miocene |
| 120–125 170–171 | 30 km to N. of Zapala on the road to Las Lajas | 1·74 | 30·31 | 54·2 | 8·6 | ±3·5 | *R* | II | Ur. Pliocene | Ur. Miocene |



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## Table 3

*Magnetic Polarities and Radiometric Ages of lavas studied in the Buta Ranquil Area, Neuquén Province*

| Sample No. | Locality | K–Ar Method: K per cent | $Ar^{40}$ rad mol. $x$ $10^{-11}$ $g^{-1}$ | $Ar^{40}$ atm. per cent | Age My | Error My | Magnetic polarity | Previous geological information: Basalt No. | Age | New classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 266–275 | Buta Ranquil | 1·90 | 1·31 | 94·9 | 0·3(3) | ±0·2 | *N* | V | Lr. Holocene | Ur. Pleistocene |
| 234–236 | Tricao Malal | 0·92 | 2·30 | 85·1 | 1·0 | ±0·2 | *R* | V | Lr. Holocene | M. Pleistocene |
| 287–288 | 9 km before Ranquil Norte, Mendoza | 1·17 | 0·50 | 93·4 | 1·3 | ±0·8 | *N* | V | Lr. Holocene | Pleistocene |
| 222–227 | Between Chos Malal and Tricao Malal, N. of the former | 2·01 | 8·89 | 88·2 | 2·3 | ±0·5 | *N* | III | Lr. Pleistocene | Lr. Pleistocene |
| | | 1·92 | 0·73 | 91·9 | 2·1 | ±0·5 | *N* | III | Lr. Pleistocene | to Ur. Pliocene |
| | Average data | | | | 2·2 | ±0·12 | | | | |
| 228–233 | Between Chos Malal and Tricao Malal, N. of the former | 1·82 | 1·76 | 85·0 | 2·5 | ±0·3 | *R* | V | Lr. Holocene | Ur. Pliocene |
| 280–281 | 1·4 km N. of R. Barrancas, on route No. 40, Mendoza | 0·66 | 4·9 | 94·6 | 3·2 | ±1·0 | *N* | IV | Ur. Pleistocene | Ur. Pliocene |

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## Table 4

*Magnetic Polarities and Radiometric Ages of lavas studied in the Los Volcanes Area, Mendoza Province*

| Sample No. | Locality | K–Ar Method: K per cent | $Ar^{40}$ rad mol. $x$ $10^{-11}$ $g^{-1}$ | $Ar^{40}$ atm. per cent | Magnetic Age My | Error My | Magnetic polarity | Previous geological information: Basalt No. | Age | New classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 383–384 | In front of Mallines de la Puntilla, on route No. 40 | 1·52 | 0·66 | 95·5 | 0·2 | ±0·1 | *N* | VI | Ur. Holocene | Ur. Pleistocene |
| 385–388 | 4 km S. of above | 1·42 | 2·26 | 95·8 | 0·55 | ±0·2 | *N* | V | Lr. Holocene | Ur. Pleistocene |
| 379–382 | 3 km N. of Mallines de la Puntilla, on route No. 40 | 0·71 | 0·28 | 98·8 | 1·5 | ±1·0 | *N* | IV | Ur. Pleistocene | Pleistocene (?) |
| 352–355 | 4 km N. of El Manzano | 0·87 | 1·04 | 94·5 | 1·8 | ±0·6 | *R* | III | Lr. Pleistocene | Lr. Pleistocene (?) |
| 375–376 | Puntilla del Huincan on route No. 40 | 0·96 | 2·26 | 88·5 | 2·6 | ±1·2 | *N* | IV | Ur. Pleistocene | Lr. Pleistocene to Ur. Pliocene |
| 361–362 | 16 km S. of Manzano on route No. 40 | 1·55 | 38·04 | 92·5 | 13·3 | ±4·0 | *N* | I | Miocene | Lr. Miocene |
| 356–358 | 8 km S. of El Manzano on route No. 40 | 1·73 | 70·79 | 34·0 | 19·4 | ±0·7 | *R* | I | Miocene | Lr. Miocene |
| | 8 km S. of El Manzano on route No. 40 | 1·69 | 47·55 | 41·5 | 27·6 | ±0·8 | *R* | I | Miocene | Lr. Miocene |



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**Table 5**

*Magnetic Polarities and Radiometric Ages of lavas studied in the Llancanello Area, Mendoza Province*

| Sample No. | Locality | K–Ar Method: K per cent | $Ar^{40}$ rad mol. $\times 10^{-11}$ $g^{-1}$ | $Ar^{40}$ atm. per cent | Age My | Error My | Magnetic polarity | Previous geological information: Basalt No. | Previous geological information: Age | New classification |
|---|---|---|---|---|---|---|---|---|---|---|
| 309–312 | Estancia Llancanello | 0·73 | 0·16 | 99·5 | 0·0–0·1 | | *N* | VIII | Ur. Holocene | Holocene–Ur. Pleistocene |
| 313–318 | Puesto Herrera, W. of Piedras Blanca | 0·63 | 0·66 | 97·2 | 0·45 | ±0·2 | *N* | III | Lr. Pleistocene | Ur. Pleistocene |
| 327–328 | S. of Las Chacras | 0·92 | 0·82 | 96·0 | 0·5 | ±0·2 | *N* | V | Lr. Holocene | Ur. Pleistocene |
| LB 5* | S. de la Ventana, Agua Botada, Malargüe | 0·80 | 16·77 | 73·7 | 6·7 | ±0·5 | | I | Miocene | Lr. Pliocene–Ur. Miocene |
| LB 10* | Ao. Agua Botada, S. of yacimiento Huemul | 0·75 | 32·34 | 69·6 | 14·4 | ±0·7 | | | Oligocene | M. Miocene |
| LB 12* | Co. Huemul, yacimiento Huemul, Malargüe | 0·68 | 31·26 | 70·9 | 17·3 | ±0·8 | | | Oligocene | M. Miocene |

* These are basic andesites belonging to the Molle Formation.

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are catalogued as Basalt II and with reversed TRM. The combined palaeomagnetic-radiometric analysis suggests that these two flows may belong to the same polarity interval.

It is noted that the basalts exposed all the way along the branch road leading to the Estancia Llamuco were catalogued in an excellent regional geological study (Lambert 1956) as Basalt III. Later, more detailed geological studies (Rodrigo 1957) indicated that the part of these flows exposed nearby road number 22 is older belonging to Basalt II while the remainder should continue to be considered as Basalt III. Valencio & Creer (1968) confirmed the correctness of this subdivision by demonstrating that the said basalts had TRM of opposed polarity (see also preceding paper); in their part, the radiometric ages presented in this paper corroborate all that has been described above. The Basalt III exposed some 2 km to the east of road number 22 is $2{\cdot}6 \pm 0{\cdot}1$ My old and the basalt exposed at the Ramadilla puesto is $6{\cdot}6 \pm 0{\cdot}3$ My old. These two dates suggest the possible occurrence of two distinct effusions later than the Basalt II.

This case provides an example of the problems to be faced in elucidating the discrepancies between stratigraphic, radiometric and palaeomagnetic ages. Nevertheless it is clear that the radiometric age of the supposed Basalt III of Ramadilla puesto is much greater than the previously assigned age, as is also the case for radiometric ages measured for other Basalt III flows exposed a little further south ($2{\cdot}3 \pm 0{\cdot}3$ My).

The lavas exposed along the road that joins Las Lajas in the north with Estancia Llamuco in the south had been catalogued as Basalt V (Lr. Holocene) by Lambert (1956): these lavas possess a reversed TRM and this can be interpreted as indicating that their age must be greater than 0·7 m.y. (Valencio & Creer 1968). Rodrigo (verbal communication) maintains that these lavas belong to Basalt IV which would be in better agreement with the minimum age suggested by the palaeomagnetic study. On the other hand, the geochronological study of these lavas indicates an age of $1{\cdot}3 \pm 0{\cdot}4$ My. Although the error is rather large, the result agrees with the age deduced from the palaeomagnetism.

Last, the lavas collected from Portada de Covunco and the Estancio Llamuco, catalogued as Basalt V (Lambert 1956; Rodrigo 1957) both have normal TRM and have radiometric ages of $0{\cdot}7 \pm 0{\cdot}2$ My and $0{\cdot}5 \pm 0{\cdot}3$ My respectively. These results are in agreement with the geological ages, and indicate that these lavas are the youngest of all those studied in the Zapala area, and together with the Basalt IV from the south of Las Lajas are the only ones which belong to the Quaternary.

### 3.3. *Buta Ranquil area*

The ages obtained from the geochronological study of the basalts from this area are summarized in Table 3. The radiometric ages are greater than the ages assigned from previous studies.

On the basis of palaeomagnetic studies, Creer & Valencio (1969), indicated that the Basalt V flows exposed in Tricao Malal and along the road from there to Chos Malal (reversed polarity) are not of the same age as those Basalt V flows exposed at Buta Ranquil (normal polarity). The former must be older than 0·7 My (Brunhes–Matuyama boundary). This has been confirmed by the K–Ar measurements: for the reversed flows we obtain ages of $1{\cdot}0 \pm 0{\cdot}2$ My and $2{\cdot}5 \pm 0{\cdot}3$ My respectively and for the normal flows $0{\cdot}3 \pm 0{\cdot}1$ My. Thus, of the flows mapped as Basalt V, those with reversed TRM must be recatalogued: also those exposed at Tricao Malal belong to a different effusive period from those exposed on the road to Chos Malal.

It is interesting to note in this respect that the flows mapped as Basalt III exposed along this road have normal TRM and a radiometric age of $2{\cdot}2 \pm 0{\cdot}4$ My.

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The flows exposed just to the north of the bridge across the R. Barrancas on R.N. 40 cannot be Basalt IV as previously supposed as their age has been determined as $3{\cdot}2 \pm 1{\cdot}0$ My.

### 3.4. *Los Volcanes area*

The radiometric ages obtained for the flows exposed in this area are in general agreement with those assigned from geological studies, although, as can be seen in Table 4 tend to be systematically slightly greater.

The oldest of the lavas studied are the flows of Basalt I exposed to the south of the Manzano stream on the west side of the R. Grande. There lavas are encountered with reversed polarity to the north of the Luchanquil stream which we have dated at 23·0 My and to the south of this stream the flows which are of normal polarity have been dated at $13{\cdot}3 \pm 4{\cdot}0$ My.

These lavas exposed on the eastern bank of the R. Grande at Puntilla de Huincan are mainly of Basalt VI (383–388). However, sample 375 and 376 from one of the sampling sites of this area have a radiometric age of $2{\cdot}6 \pm 1{\cdot}2$ My. Originally Valencio & Creer (1968) thought that these samples like 383–388 should correspond to Basalt VI since all have normal TRM. However, the radiometric age obtained for them is much older than those obtained for the other Basalt VI samples from this area (383–384) ($0{\cdot}2 \pm 0{\cdot}1$ My). This difference in age is too great to be attributed to experimental error. In fact, a relict of Basalt IV has been reported in this area immediately to the east of a narrow belt of Basalt VI (Groeber 1937) and we conclude that the exposure from which samples 375 and 376 were collected lies in this relict of Basalt IV.

The youngest basalts are classified as Basalt V ($0{\cdot}6 \pm 0{\cdot}2$ My) and Basalt VI ($0{\cdot}2 \pm 0{\cdot}1$ My), both having normal TRM. These radiometric ages indicate they are not Holocene as postulated in previous geological studies.

### 3.5. *Llancanelo area*

The radiometric ages obtained are summarized in Table 5, where data for Basalt I flows and for basic andesites from the vicinity of Agua Botada stream to the south of Malargüe are also included.

The oldest of the lavas studied are the basic andesites of Cerro Huemul and Agua Botada stream. These are considered to belong to the Molle formation (Oligocene). However, the ages obtained are $17{\cdot}3 \pm 0{\cdot}8$ and $14{\cdot}4 \pm 1{\cdot}0$ My respectively which correspond to the Middle Miocene and not to the Oligocene. It is therefore considered desirable that a revision of previous stratigraphic work be carried out in order to test our conclusion.

The youngest lavas in this area are of Basalt VII nearby Llamuco Estancia. The accuracy of the age determined (0·0 to 0·1 My which is strongly affected by atmospheric argon contamination) places them in the Holocene or Upper Pleistocene. Of all the flows studied and mapped as Holocene (Basalts V, VI and VII) in the different areas, these are the only ones whose radiometric age confirms that they could belong to this epoch.

### 3.6. *General discussion*

The results summarized in Tables 1–5 are collected together and presented in Fig. 1, where it may be observed that the lava flows studied have been divided into two groups to facilitate the analysis; viz: (1) central and south Neuquén extending from Las Lajas in the north to the 40° S parallel of latitude and (2) north Neuquén and south Mendoza extending from Chos Malal in the south to Malargue in the north. The two zones are separated by an extensive area between Las Lajas and

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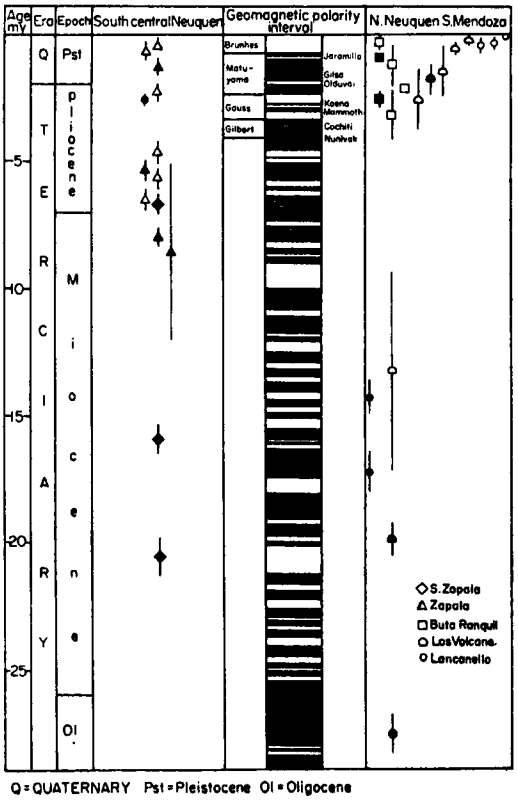



FIG. 1. Radiometric ages with estimated errors. Sampling area indicated by the shape of the symbol (see Key). Open symbols indicate normal magnetic polarity; black symbols indicate reversed magnetic polarity.

Chos Malal in which samples were not collected. The following conclusions are drawn:

(i) In zone (2) a magmatic cycle beginning in the Upper Pliocene and lasting until the end of the Pleistocene can be defined. At various times within this interval were erupted the majority of the Cenozoic basalts studied.

(ii) This magmatic cycle may also be recognized in zone (1). However, in the south of zone (1) the youngest flows studied are of Basalt V. It is planned to study Basalts VI and VII from this zone in the future.

(iii) In zone (1) a second magmatic cycle beginning in the Upper Miocene and lasting until the Middle Pliocene may be defined. This cycle cannot be recognised in zone (2) from which only the Basalt I flows from Sierra de la Ventana in Malargue are of the same age. But further studies are desirable to enquire whether the absence of such dates is due to chance.

(iv) It is not possible to recognize any grouping of dates within the Miocene.

## 4. The bearing of the radiometric results on the magnetic stratigraphy

### 4.1. *On combined radiometric–palaeomagnetic studies*

The errors we have attached to the radiometric data presented here are larger than those usually assigned in similar determinations. As explained in Section 2.3

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this is because we quote the error due to atmospheric argon 40 contamination rather than the simple experimental error which is approximately 3 per cent.

In Fig. 1 the time scale of reversals of the geomagnetic field during the most recent 4 My, compiled from the recent review by Cox, Doell & Dalrymple (1968) is illustrated. Also part of the hypothetical time scale for earlier time (Heirtzler *et al.* 1968) is shown. Logically, we should only include in the comparison those samples whose ages are subject to small errors, of the order of magnitude of the magnetic event or epoch to which they belong. This would, however, restrict us to using only the younger data and in fact we have illustrated all our data in order to bring out the problems involved in extending combined palaeomagnetic–radiometric studies beyond about 4 My ago.

### 4.2. *The Brunhes epoch* (0·0–0·7 *My*)

The normal polarity of the epoch is illustrated by the results from Basalt V from Buta Ranquil, from Basalt VI from Los Volcanes and from Basalts III, V and VII from Llancanelo. The known lower boundary of this epoch at 0·7 My allows us to determine the ages of (a) Basalt V from Llamuco Estancia (Zapala area), (b) Basalt V from the Portada de Covunco (Zapala area) and (c) Basalt V from the Los Volcanes area with greater precision than indicated by the estimated errors in the K–Ar dates listed in the Tables.

### 4.3. *Boundary between the Matuyama and Gauss epochs*

Cox & Dalrymple (1967) describe a statistical method of defininining the most probable ages of polarity transitions of the geomagnetic field, making use of all the relevant data. They deduced that the Gauss–Matuyama transition occurred 2·5 My ago.

A list of dates of about this age measured for our Argentine samples is given in Table 6. Cox & Dalrymple (1967), in their analysis, accepted only dates with estimated precisions of less than 5 per cent and rejected the others. We have quoted errors estimated due to *argon contamination* (see Section 4.1) whereas *instrumental* errors are usually quoted. The latter are less than 5 per cent for the first four entries in Table 6 and thus all of these might be considered acceptable for inclusion in a new statistical analysis. The result from Buta Ranquil, age $2{\cdot}2 \pm 0{\cdot}1$ My with normal polarity, may correspond to a recently discovered short normal event at about 2·2 My within the Matuyama epoch (Cox 1969).

### 4·4. *Discussion of ages greater than* 4 *My.*

So far palaeomagnetic and radiometric studies have resulted in the accurate construction of the stratigraphy of geomagnetic reversals to 4 My ago. Various authors have attempted to extend the column of reversals to earlier time from interpretations of magnetic anomaly patterns over the ocean floors. By this means it is possible to observe the record of reversals as far back as 70 My ago, but the time scale is arbitrary, depending on the assumption that the rate of sea-floor spreading has, on the whole, been that established for the most recent 4 My by using the accumulated palaeomagnetic and potassium–argon results obtained from studies of lavas exposed on land. It is thus desirable to use suitable results from the older continental basalts to provide a control of the arbitrary time scale of the reversal pattern deduced from sea-floor spreading. It is obviously futile to attempt to do this for lavas more than about 10 My old because the errors with which the age can be determined radiometrically become longer than the length of an interval of fixed

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geomagnetic polarity. Nevertheless, it is possible that the accumulated results from rocks up to about 10 My old may eventually verify the extrapolation about sea-floor spreading rates used in drawing up the extended sequence of reversals.

**Table 6**

| Area | Sample Nos. | See Table No. | Age My | Polarity | Basalt No. |
|---|---|---|---|---|---|
| Zapala | 141–144 | 2 | 2·3±0·3 | *N* | III |
| Zapala | LB 37 | 2 | 2·6±0·1 | *N* | III |
| Buta Ranquil | 222–227 | 3 | 2·2±0·1 | *N* | III |
| Buta Ranquil | 228–233 | 3 | 2·5±0·3 | *R* | V |
| Los Volcanes | 375–376 | 4 | 2·6±1·2 | *N* | IV |

From our present study the results below are considered of significance.

(a) The following flows have reversed TRM and can be correlated with periods of reversed polarity (see Fig. 1):

(i) Basalt I from the south of Las Lajas in the Zapala area (5·4±0·3 My);

(ii) Basalt II from south of Catan Lil in the south Zapala area (6·7±0·3 My); and

(iii) Basalt II from the south of the city of Zapala (8·0±0·2 My).

(b) Basalt I from south of Las Lajas has reversed polarity (5·4±0·3 My) while that from south of Zapala has normal polarity (5·7±0·4 My). We expect to find the precise position of this R/N boundary (which we call G1) when we have obtained more and better radiometric dates for these basalts. An R/N boundary occurs at approximately 5·7 My on the hypothetical polarity reversal column deduced from sea-floor spreading (Fig. 1).

(c) We designate as G2 the N/R boundary given by the normally magnetized Basalt III from Puesto Ramadilla in the Zapala area (6·6±0·3 My) and the reversely magnetized Basalt II from south of Catan Lil in the south Zapala area (6·7±0·3 My). G2 may be R/N rather than N/R because of the uncertainties of the measured ages. We note that there is a narrow interval of reversed polarity about here in the hypothetical reversal column.

We have given the names G1 and G2 to these transitions to commemorate the work of Dr Pablo Groeber, one of the pioneers of geological studies in Argentina especially on the stratigraphy of the lavas we have studied.

### 4.5. *Geomagnetic epochs and events*

In the past few years it has become established practice to distinguish polarity 'events' within longer polarity 'epochs'. Reference to Fig. 1 however, indicates that such a distinction cannot reasonably be made between the polarity time intervals found for the past 12 My. There do not appear to be two distinguishably different characteristic times during which the field maintained one particular polarity, but rather a continuous spectrum of times. Historically, a distinction was drawn between 'epochs' and 'events' partly because of the naturally greater difficulty in detecting and then proving the existence of the shorter polarity intervals (the so-called events). 'Epochs' were first discovered and some of these were later found to contain 'events'. However, the distinction is artificial and somewhat misleading in that it implies that there are two kinds of geomagnetic field reversal. While magnetohydrodynamic theories of origin of the geomagnetic field allow reversals

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of polarity of the main dipole, they offer no reasons why we should expect to find two time scales of reversal. They predict rather a whole spectrum of times scales and this is in agreement with the experimental evidence for the past 70 My deduced from sea-floor spreading.

## 5. Conclusions

(i) In all areas except Llancanelo, the radiometric ages of these Cenozoic basalts are systematically older than the previously assigned geological ages. Hence it appears necessary to revise the stratigraphy of some areas, as indicated in Tables 1–5.

(ii) The extra-Andean zone extending from the south of Mendoza province to the south of Neuquén province experienced a magmatic cycle which began in the upper Pliocene and lasted until the end of the Pleistocene. The radiometric ages obtained do not suggest the occurrence of effusions in the Holocene, except perhaps in Llancanelo.

(iii) A second magmatic cycle has been defined in the centre and south of Neuquen province between the Upper Miocene and Middle Pliocene.

(iv) The R/N boundary between the Matuyama and Gauss epochs may be rather more recent than the age of 2·5 My at present accepted, or alternatively we may have detected a short normal event at about 2·2 My within the Matuyama.

(v) Two reversals of polarity G1 and G2 have been dated at $5{\cdot}5 \pm 0{\cdot}3$ My (R/N) and at $6{\cdot}65 \pm 0{\cdot}3$ My (N/R?).

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## References

Amaral, G. *et al.*, 1966. Potassium–argon dates of basaltic rocks from southern Brazil, *Geochim. Cosmochim. Acta*, **30**, 159–189.

Cordani, U. G., 1968. Idade do volcanismo no Oceano Atlantico Sul. Tesis, University São Paulo, Brazil (unpublished).

Cox, A., 1969. Geomagnetic Reversals. *Science*, **163**, 237–245.

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Cox, A. & Dalrymple, G. B., 1967. Statistical analysis of geomagnetic reversal data and the precision of potassium–argon dating, *J. geophys. Res.*, **72,** 2603–2614.

Cox, A., Doell, R. R. & Dalrymple, G. B., 1968. Radiometric time-scale for geomagnetic reversals, *Q. Jl. R. astr. Soc.*, **124,** 53–66.

Creer, K. M. & Valencio, D. A., 1969. Palaeomagnetic and rock magnetic studies on the Cenozoic basalts from western Argentina, *Geophys. J. R. astr. Soc.*, in press.

Damon, P. E., 1966. Correlation and chronology of ore deposits and volcanic rocks, Ann. Rep. C00–688–60. Geochronology Lab., University of Arizona, Tucson, Arizona.

Groeber, P., 1937. Hoja 30c del Mapa Geologico general de la Republica Argentina, Mapa sin texto; Direccion de Minas y Geoloiga.

Heirtzler, J. R., Dickson, G. O., Herron, E. M., Pitman, W. C. & Le Pichon, X., 1968. Marine magnetic anomalies, geomagnetic field reversals and motions of the ocean floor and continents, 1968, *J. geophys. Res.*, **73,** 2119–2136.

Lambert, L. R., 1956. Descripcion geologica de la hoja 35b, Zapala (T. N. de Neuquen), *Direc. Nac. Miner.*, Bol. 83.

Reynolds, J. H., 1956. High sensitivity mass spectrometer for noble gas analysis, *Rev. Sci. Inst.*, **27,** 928–934.

Rodrigo, F., 1957. Informe geologico de la hoja I.G.M. 1 : 100·000, 3969–13, Zapala, *Dir. Gral. Ingenieros* (inedito)

Valencio, D. A. & Creer, K. M., 1968. El paleomagnetismo de algunas lavas Cenozoicas de la Republica Argentina, *Rev. Asoc. Geol. Arg.*, **XXIII,** 4.

## APPENDIX

### Petrographic description of the samples studied

*Sample LB*–5. Basalt I. Sierra de la Ventana, Malargüe, Mendoza. *Olivine basalt.* Porphyritic texture composed of abundant hipidiomorphic phenocrysts of olivine, plagioclase and clinopyroxene in a holocrystalline groundmass of plagioclase, clinopyroxene, olivine and magnetite (?). The phenocrysts constitute more than 50 per cent of the rock.

*Sample LB*–10. Agua Botada stream to the south of Huemul ore deposit, Malargüe, Mendoza. *Basic indesite.* Porphyritic texture with large phenocrysts of clinopyroxene with borders surrounded by plagioclase distributed in a groundmass composed of altered plagioclase grains, chlorite, pyroxene and magnetite.

*Sample LB*–12. Cerro Huemul. Huemul ore deposit, Malargüe, Mendoza. *Basalt.* Porphyritic texture with large phenocrysts of clinopyroxene and plagioclase, distributed in a groundmass composed of platelets of plagioclase, pyroxene, chlorite, epidote and opaques.

*Sample LB*–37. Basalt III. Road to Llamuco Estancia, 2 km to the west of road No. 22, Zapala, Neuquén. *Olivine basalt.* Porphyritic texture with olivine phenocrysts, altered to iddingsite along the edges, and plagioclase phenocrysts distributed in a fine grained groundmass of platelets of plagioclase, olivine, abundant opaques and possible vitreous mesostasis.

*Sample* 105. Basalt V. Portada Covunco to the south of Zapala Neuquén. *Olivine basalt.* Hipidiomorphic phenocrosts of pyroxene and olivine, in part corroded, in an integranular groundmass composed of pyroxene.



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*Sample* 122. Basalt II. 30 km to the north of Zapala, road to Las Lajas, Neuquén. *Olivine basalt.* Porphyritic texture with idiomorphic phenocrysts of olivine, partly altered to iddingsite, distributed in a groundmass composed of plagioclase, clinopyroxenes and opaques.

*Sample* 129. Basalt III. Ramadilla village. Estancia Llamuco, Zapala, Neuquén. *Olivine basalt.* Porphyritic texture with abundant olivine phenocrysts, in a groundmass composed of thin platelets of plagioclase forming a trachytic texture, situated among pyroxene and opaque grains.

*Sample* 142. Basalt III Puesto Ramadilla, Llamuco Estancia, Zapala, Neuquén. *Basalt.* Porphyritic texture with hipidomorphic plagioclase and pyroxene phenocrysts in a groundmass consisting principally of pyroxenes and opaques.

*Sample* 148. Basalt II, 5 km to the south of Zapala, Neuquén. *Olivine basalt.* Plagioclase and olivine phenocrysts in an intergranular groundmass composed of pyroxene and plagioclase.

*Sample* 153. Basalt I. 13 km south of Zapala, on route No. 40, Neuquén. *Olivine basalt.* Porphyritic texture weakly evident with olivine and pyroxene phenocrysts, in a groundmass composed of plagioclase platelets, pyroxene and opaques together with secondary carbonate.

*Sample* 154. Basalt I. 113 km south of Zapala on route No. 40, Neuquén. *Olivine basalt.* Porphyritic texture with abundant olivine and clinopyroxene phenocrysts in an intergranular groundmass composed of plagioclase, pyroxene and opaques.

*Sample* 163. Basalt III. 25 km south of Catan Lil, Neuquén. *Basalt.* Porphyritic texture with idiomorphic ferromagnesian phenocrysts, almost totally replaced and abundant amygdales filled with carbonate, distributed in a hypocrystalline groundmass with platelets of plagioclase and hipidomorphic grains of pyroxene.

*Sample* 166. Basalt II. 37 km south of Catan Lil, Neuquén. *Olivine basalt.* Large idiomorphic phenocrysts of olivine with corroded borders and abundant inclusions of plagioclase in a groundmass consisting of the same components.

*Sample* 177. Basalt I. 20 km to the north of Las Lajas, Neuquén. *Olivine basalt.* Porphyritic texture with xenomorphic pyroxene and hipidiomorphic plagioclase, and idiomorphic olivine phenocrysts distributed in a sub-ophitic groundmass of plagioclase and pyroxene together with an opaque.

*Sample* 178. Basalt I. 16 km to the south of Las Lajas, Neuquén. *Olivine basalt.* Abundant small phenocrysts of idiomorphic plagioclase and large olivine phenocrysts in a fine grained groundmass composed of plagioclase, pyroxene and opaques.

*Sample* 204. Basalt IV. 35 km to the south of Las Lajas, Neuquén. *Olivine basalt.* Porphyritic texture with abundant altered plagioclase phenocrysts of various sizes, corroded and altered phenocrysts of olivine and clinopyroxene. The groundmass is composed of abundant plagioclase grains of sub-quadratic shapes accompanied by clinopyroxene and an opaque.

*Sample* 222. Basalt III. Between Chos Malal and Tricao Mala, Neuquén. *Basic andesite or basalt.* Porphyritic texture with a large quantity of plagioclase phenocrysts and some of pyroxene. The groundmass is hyalopilitic with microlites of plagioclase, pyroxene and an opaque.

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*Sample* 225. Basalt III. Same locality as sample 222. *Olivine basalt.* Large and abundant phenocrysts of plagioclase of wide plate-like shape and some smaller ones of olivine in a groundmass of plagioclase, pyroxene and opaques.

*Sample* 229. Basalt V. To the north of Chos Malal, Neuquén. *Basalt.* Porphyritic texture with abundant homogeneously oriented and distributed plagioclase phenocrysts and clinopyroxene phenocrysts. Very small platelets of plagioclase and a minor proportion of pyroxene and opaques are distributed in the groundmass.

*Sample* 234. Basalt V. Tricao Malal, Neuquén. *Olivine basalt.* Porphyritic texture with a large abundance of plagioclase phenocrysts of various sizes and of hipidiomorphic partly corroded and altered olivine phenocrysts, The fine grained groundmass is of intergranular texture composed of plagioclase and pyroxene.

*Sample* 273. Basalt V. Buta Ranquil, Neuquén. *Olivine basalt.* Porphyritic texture with a large quantity of tabular and subquadratic plagioclase phenocrysts and a smaller quantity of clinopyroxene. The groundmass is composed principally of microlites of plagioclase in a vitreous base.

*Sample* 280. Basalt IV. 1·4 km to the north of R. Barrancas, on route No. 40, Mendoza. *Olivine basalt.* Porphyritic texture with abundant idiomorphic olivine phenocrysts with plagioclase inclusions in an intergranular groundmass of plagioclase, pyroxene and opaques.

*Sample* 287. Basalt V. 9 km before North Ranquil, Mendoza. *Olivine basalt.* Porphyritic texture with large idiomorphic partly altered olivine phenocrysts distributed in a groundmass of intergranular texture of plagioclase platelets and abundant pyroxene grains.

*Sample* 309. Basalt VII, Llancanelo Estancia, Mendoza. *Olivine basalt.* Hipidiomorphic phenocrysts of plagioclase and of olivine (abundant) in an intergranular groundmass composed principally of plagioclase platelets and a minor proportion of pyroxene.

*Sample* 313. Basalt III. Puesto Herrera, to the west of Piedras Blancas, Mendoza. *Olivine basalt.* Porphyritic texture with large idiomorphic phenocryst of olivine in a groundmass, partly of subophitic texture of elongated platelets of plagioclase and of pyroxene grains.

*Sample* 327. Basalt V. To the south of Las Chacras, Mendoza. *Olivine basalt.* Porphyritic texture with hipidiomorphic olivine phenocrysts and idiomorphic plagioclase phenocrysts distributed in an intergranular groundmass composed of plagioclase platelets, pyroxene grains and iron oxides.

*Sample* 352. Basalt III. To the north of El Manzano, Mendoza. *Olivine basalt.* Porphyritic texture with small rounded olivine crystals and tabular plagioclase in a groundmass containing fine grained plagioclase with chlorite mesostasis, carbonates and opaques.

*Sample* 358. Basalt I. 8 km to the south of El Manzano, Mendoza. *Olivine basalt.* Porphyritic texture with large hipidiomorphic phenocrysts of olivine, with edges altered to iron oxide, epidote and iddingsite distributed in an intergranular groundmass of plagioclase, pyroxene, epidote and opaques.

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*Sample* 362. Basalt II, 16 km to the south of El Manzano, Mendoza. *Basic andesite.* Porphyritic texture with abundant zoned plagioclase phenocrysts, included olivine partly altered to iddingsite distributed in a groundmass of plagioclase with pilotaxic appearance and pyroxene grains.

*Sample* 375. Basalt IV. Puntilla de Huincan, Mendoza. *Basalt.* Porphyritic texture with large plagioclase phenocrysts some of which are sub-quadratic and a smaller proportion of olivine phenocrysts in a fine grained groundmass composed of plagioclase platelets and abundant pyroxene grains and opaque grains.

*Sample* 379. Basalt IV. 3 km to the north of Los Mallines de la Puntilla, Mendoza. *Olivine basalt.* Porphyritic texture with abundant olivine phenocrysts with corona structure and included pyroxene phenocrysts and a large proportion of plagioclase of grain sizes varying from phenocryst to microlite.

*Sample* 385. Basalt V. 4 km south of Los Mallines de la Puntilla, Mendoza. *Olivine basalt.* Porphyritic texture with abundant homogeneously orientated olivine phenocrysts with highly corroded borders and included sub-quadratic plagioclase phenocrysts. The groundmass consists of a dense mass of plagioclase platelets between which pyroxene grains are situated in intergranular texture and also of opaque grains.

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